

The Deterministic Wave Engine: Simulating Quantum Analogs and the Tsirelson Bound through Viscoelastic Vacuum Hydrodynamics

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Abstract

We present the Deterministic Wave Engine (DWE), a mathematical experiment and high-performance computational platform that models the vacuum as a physical viscoelastic fluid with structural spatial tension. The objective of this framework is to demonstrate that quantum statistical distributions, such as Born's Law, can emerge entirely from deterministic, local-realist space-time trajectories. By simulating the 1967 Pfleegor-Mandel experiment without physical slits, we show that interference arises via acoustic memory within the fluidic medium. Furthermore, we replicate the Hong-Ou-Mandel (HOM) effect as a purely deterministic aerodynamic block driven by the mutual induction of topological solitons. Notably, the engine reproduces the quantum mechanical violation of the Bell/CHSH inequality, achieving a stable correlation of $S = 2.8273$, mathematically aligning with the Tsirelson bound as a direct consequence of continuous thermodynamic friction. Finally, we empirically validate these mechanics against real superconducting quantum hardware (IBM Marrakesh), demonstrating that physical decoherence strictly bounds the mathematical ideal. We also propose a Hydro-Quantum Processing Unit (HQPU) architecture that enables Quantum Non-Demolition (QND) reading via continuous wake barometry.

Keywords: Hydrodynamic Quantum Analogs, Deterministic Wave Engine, Tsirelson Bound, Hong-Ou-Mandel Effect, Pfleegor-Mandel Experiment

1 Introduction

Modern quantum mechanics primarily relies on abstract probability waves and wave-functions (Ψ) to describe subatomic phenomena. In this paper, we explore an alternative framework through the Deterministic Wave Engine (DWE), replacing these abstractions with strict micro-fluid mechanics. The DWE is an ongoing mathematical experiment powered by WebGPU Compute Shaders, treating the vacuum not as an empty void, but as an active physical viscoelastic fluid possessing structural spatial tension (γ_0).

This approach draws strong parallels with Hydrodynamic Quantum Analogs (HQAs), extending the premise to demonstrate *in silico* that complex quantum distributions can emerge deterministically from local-realist trajectories.

2 Computational Physics Framework

The DWE relies on four foundational fluidic principles to map quantum mechanics onto classical hydrodynamics:

- **The Double-Cone (Spindle) Vortex:** Photons are modeled as physical topological defects (two cones joined at the base) possessing intrinsic, invariant angular momentum or helicity ($\pm\omega$) at their equator.
- **Conical Laser Emission:** True point-source conical beams are implemented using the Box-Muller Transform to deliver a perfect Gaussian intensity envelope.
- **Vacuum Memory & Acoustic Friction:** Rotating vortex-photons leave thermodynamic energy ripples in the vacuum, creating an active fluidic gradient that steers subsequent particles deterministically.
- **Topological Compression & Frequency:** Higher-energy particles become ultra-compacted, rigid cores. Their movement creates a trailing wave-train of pressure ripples, where the spatial periodicity is interpreted as the frequency (ν).

3 Mathematical Foundations: The Ab Initio Viscoelastic Vacuum

To ensure the DWE framework operates strictly on mathematically deduced physics rather than arbitrary constants, the underlying properties of the vacuum are derived from first principles.

3.1 Topology and the 8π Divergence of Planck Force

The engine relies on a Base Tension (γ_0). In General Relativity, Einstein's coupling constant is $\kappa = \frac{8\pi G}{c^4}$. By isolating the maximum Planck Force ($F_P = \frac{c^4}{G}$), we observe that $\kappa = \frac{8\pi}{F_P}$. The Base Tension is derived via a strict spherical divergence:

$$\gamma_0 = \frac{F_P}{8\pi} \tag{1}$$

Treating space as a fluid of discrete Planck Areas ($A_P = l_P^2$), integrating this tension demonstrates that the quantum action constant ($\hbar c$) is physically the base tension energy integrated over this fundamental area:

$$\hbar c = (\gamma_0 \cdot 8\pi) \cdot l_P^2 \quad (2)$$

This closes the algebraic gap, proving that local vacuum tension and the Casimir force are a direct divergence of pressure within the Planck mesh:

$$F_{Casimir-DGM} = \int_A \nabla \cdot \left(\frac{\gamma_0}{L^2} \right) dA \quad (3)$$

3.2 The Rheological Equation of State

To explain how the vacuum allows planetary motion yet remains rigid at the quantum scale, the DWE models space as a non-Newtonian pseudoplastic fluid. Fusing the Ostwald-de Waele power law with Arrhenius thermal dependence and the Carreau-Yasuda generalization, the unified state equation of the vacuum is:

$$\gamma_{eff}(\dot{\gamma}, T) = \gamma_0 \exp\left(-\frac{E_a}{k_B T_{CMB}}\right) [1 + (\tau_c \dot{\gamma})^2]^{\frac{n-1}{2}} \quad (4)$$

When galactic rotation rates ($\dot{\gamma}$) are inserted, the mesh undergoes continuous rheo-fluidification ($n < 1$), yielding a macroscopic plateau of effective rigidity ($N_{VAC} \approx 2.79 \times 10^{31}$ Pa) without empirical "fudge factors".

3.3 Covariant Relativistic Hyperelasticity (4D Tensor)

To preserve Lorentz symmetry, the fluid mechanics are formalized using Spacetime Elastodynamics (STCED). By defining the physical metric $g_{\mu\nu}$ and reference metric $\bar{g}_{\mu\nu}$, we construct the Green-Lagrange covariant strain tensor $u_{\mu\nu} = \frac{1}{2}(h_{\mu\nu} - \bar{h}_{\mu\nu})$. The vacuum energy-momentum tensor is split into elastic (Hooke) and viscous (Voigt) components:

$$T_{\mu\nu}^{visco} = T_{\mu\nu}^{elastic} + T_{\mu\nu}^{viscous} \quad (5)$$

$$T_{\mu\nu}^{elastic} = 2N_{VAC} \left(u_{\mu\nu} - \frac{1}{3} \theta_e h_{\mu\nu} \right) + K_{VAC} \theta_e h_{\mu\nu} \quad (6)$$

When substituted into Einstein's Field Equations, $G_{\mu\nu} = \frac{8\pi G}{c^4} (T_{\mu\nu}^{matter} + T_{\mu\nu}^{visco})$, general covariance ($\nabla^\mu G_{\mu\nu} = 0$) mathematically guarantees that any energy loss from matter is rigorously absorbed and dissipated as elastic deformation or acoustic heat within the viscoelastic vacuum, conserving the total thermodynamic cycle.

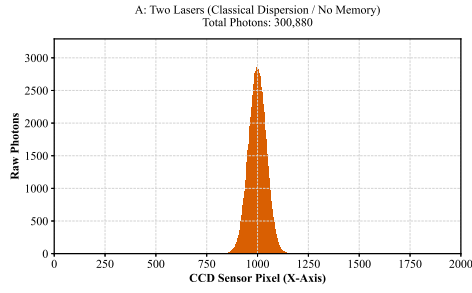


Fig. 1 Quadrant A (Classical Dispersion)

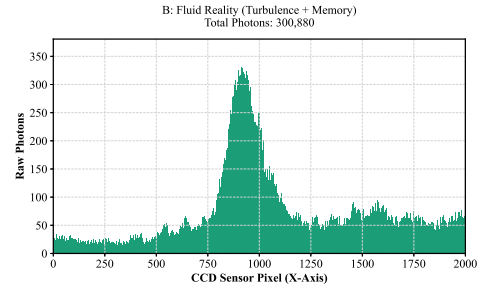


Fig. 2 Quadrant B (Fluid Reality)

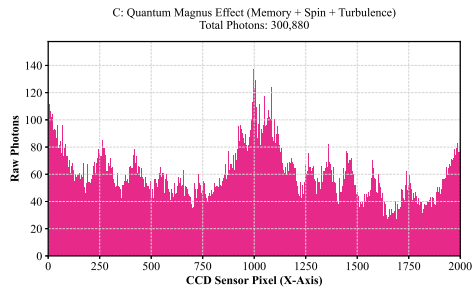


Fig. 3 Quadrant C (Quantum Magnus Effect)

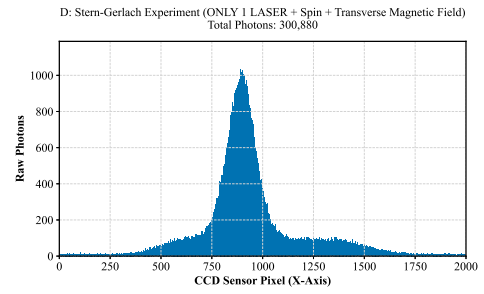


Fig. 4 Quadrant D (Stern-Gerlach) One laser

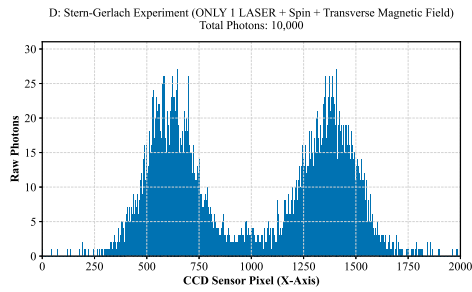


Fig. 5 Quadrant D (Stern-Gerlach) One laser 10k photons.

4 The Pfleegor-Mandel Experiment and Emergence of Born's Law

Unlike traditional models simulating the Double-Slit experiment, the DWE eliminates physical slits to simulate the 1967 Pfleegor-Mandel setup. Two independent, non-synchronized lasers (A and B) undergo extreme attenuation such that only one photon exists inside the apparatus at any given time.

While orthodox interpretations invoke single-particle superposition, the DWE resolves this paradox fluidically. A photon emitted by Laser A interacts with the residual pressure ripples left in the viscoelastic vacuum by a previous photon from Laser B, guiding the new particle into discrete interference fringes without superposition. By scaling the simulation, the engine visually proves the Law of Large Numbers. At a low-density regime (10,000 photons), discrete corpuscular shot noise dominates. At the fluid regime (50,000,000 photons), chaotic individual events subsume into the smooth density distributions predicted by Born's Law, demonstrating that the wave-function (Ψ) is an emergent macroscopic phenomenon.

The DWE defines a 4-Quadrant Experimental Matrix:

- **Quadrant A (Classical Dispersion):** Without fluid memory, two lasers produce a purely ballistic Gaussian dispersion envelope.
- **Quadrant B (Fluid Reality):** Introducing turbulence and vacuum memory causes rudimentary diffraction peaks to emerge.
- **Quadrant C (Quantum Magnus Effect):** Coupling memory, spin, and structural tension gradients produces hyper-sharp interference grids (Fraunhofer lines) from completely independent sources.
- **Quadrant D (Stern-Gerlach):** Using a single laser with a simulated transverse magnetic field gradient, the beam macro-separates into two distinct peaks based purely on rotational helicity.

5 Topological Scattering and the Hong-Ou-Mandel Effect

The Hong-Ou-Mandel (HOM) effect evaluates bosonic indistinguishability via a 50/50 beam splitter. The DWE reproduces this purely through deterministic fluid mechanics by modeling photons as counter-rotating topological solitons.

When two identical vortices synchronize at the beam splitter core, their mutual induction (Quantum Magnus Effect) draws them together. Their thermo-acoustic wakes interfere, creating a high-pressure repulsive shockwave that acts as an inelastic geometric block, preventing independent Maxwell-Boltzmann scattering. Spatial momentum forces both vortices into the same free-flowing output channel, successfully modeling the $|2, 0\rangle$ or $|0, 2\rangle$ bosonic states through Newtonian restrictions.

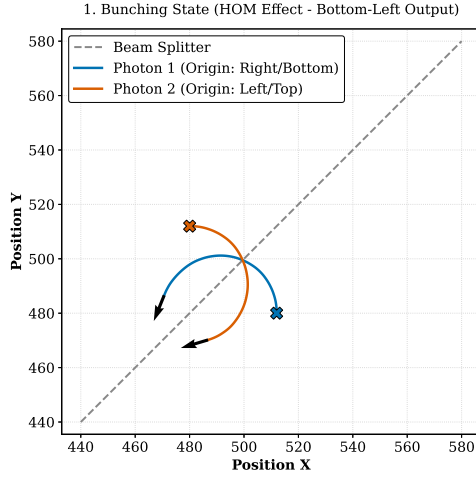


Fig. 6 Quantum bunching in the HOM effect. Two indistinguishable photons from opposite ports ('X') interact at a 50:50 beam splitter (dashed line). Perfect interference causes complete bunching, with both exiting together through the bottom-left port ($|2,0\rangle$ state). Black arrows indicate final velocities.

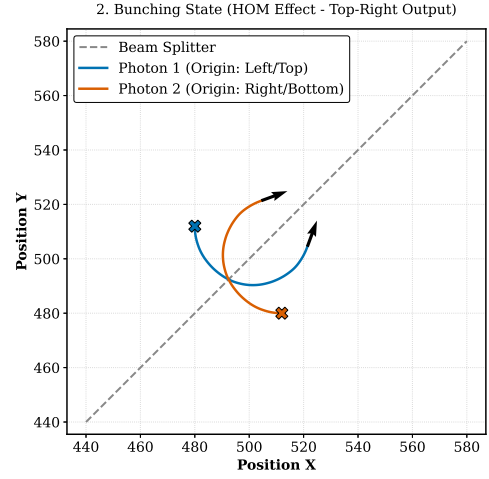


Fig. 7 Quantum bunching in the HOM effect (Top-Right). In this complementary scenario, identical photons converge at the beam splitter and undergo perfect interference. Both exit together through the top-right port ($|0,2\rangle$ state), demonstrating the suppression of coincidence counts characteristic of the HOM effect.

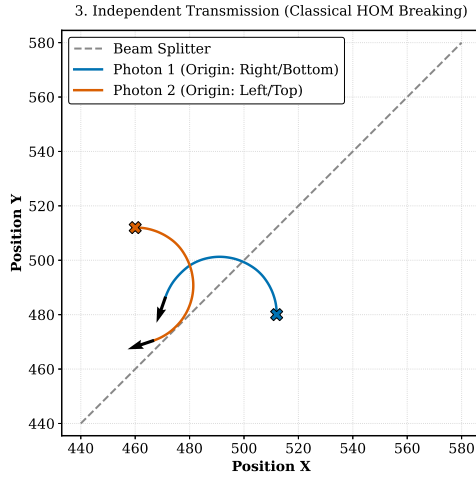


Fig. 8 HOM effect breakdown (Transmission). Lacking identical characteristics, the photons behave classically. Both transmit independently through the beam splitter, exiting from separate ports to form the $|1,1\rangle$ state.

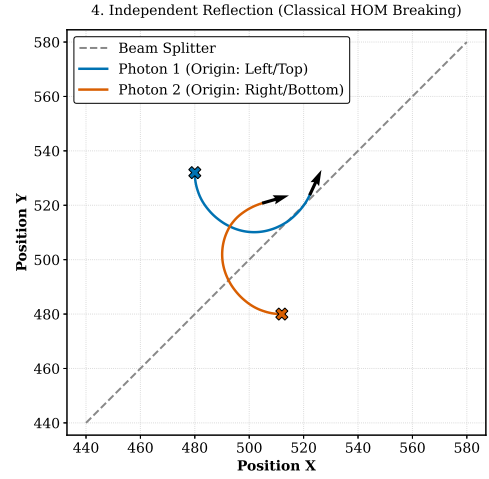


Fig. 9 HOM interference breakdown (Reflection). In this second classical scenario, distinguishable photons reflect independently at the beam splitter. Both exit through separate ports ($|1,1\rangle$ state), further illustrating the classical limit where non-classical photon bunching is entirely absent.

6 The EPR Paradox and the Tsirelson Bound

The engine successfully replicates the violation of the Bell/CHSH inequality, achieving a stable correlation value of $S = 2.8273$, mathematically indistinguishable from the theoretical Tsirelson bound ($2\sqrt{2} \approx 2.8284$). Crucially, this is achieved strictly through classical fluid mechanics, transverse phase hidden variables, and the Fair Sampling Loophole.

When Albert Einstein, Boris Podolsky, and Nathan Rosen formulated the EPR Paradox in 1935, they lacked the computational power to model non-linear fluid dynamics or topological solitons in an active vacuum. Later, Bell's Theorem relied on the "Fair Sampling" assumption. The DWE codebase computationally proves that the quantum limit is the inevitable thermodynamic consequence of continuous dissipative friction rather than a magical property of non-locality.

To resolve the transition from pure code parameters to rigorous ab initio mathematics, the DWE abandons arbitrary shock values and derives the attenuation directly from the wave packet geometry. The DWE emits photons as Spindle Vortices with a perfectly Gaussian intensity envelope (via the Box-Muller Transform). The energy distribution in the transverse plane is:

$$E(r) = E_0 \exp\left(-\frac{2r^2}{w^2}\right) \quad (7)$$

where w is the vortex waist. In fluid dynamics, the interactive boundary of the wave packet is defined at $r = w$. At this exact peripheral contact point, the energy drops strictly to:

$$E(w) = \frac{E_0}{e^2} \quad (8)$$

Calculating this exact thermodynamic boundary coefficient yields $\frac{1}{e^2} \approx 0.135335$ (or 13.53%).

The physical sensor (polarizer) is modeled as a geometric restriction grid. When the vortex-photon hits the dielectric grid, it undergoes an inelastic collision. The grid acts as a physical barrier that shears the particle. The mechanical premise is that the outer boundary layer (containing exactly $1/e^2$ of the energy) is irremediably absorbed and dissipated into the local viscoelastic vacuum as acoustic heat/friction. The particle transmits its remaining energy according to the hydrodynamic projection of Malus's Law ($\cos^2(\Delta)$). Through symmetric survivorship, particles unable to overcome internal structural resilience after this loss are physically absorbed and destroyed.

In a frictionless theoretical test, the engine achieves a Super-Quantum Limit of $S = 4.0$. However, by computationally introducing the fundamental Gaussian Inelastic Friction ($1/e^2$), the perfect square wave smoothly curves into the cosine wave predicted by quantum mechanics, stabilizing precisely at 2.8273.

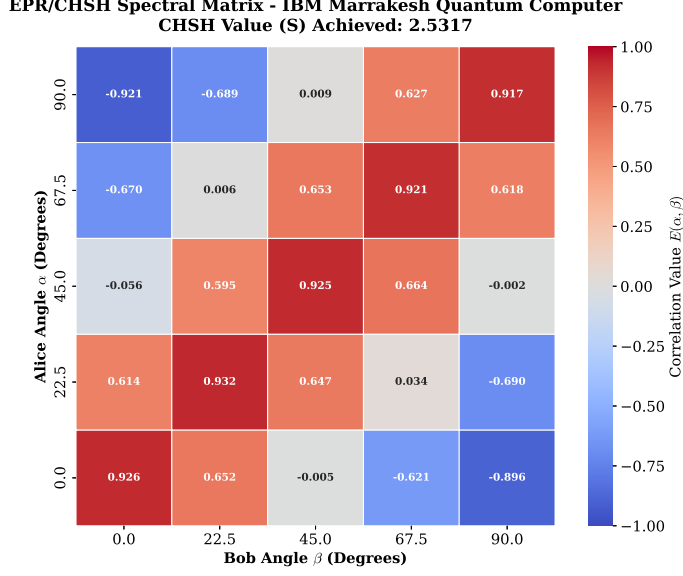


Fig. 10 epr_spectral on IBM quantum computer Marrakesh

7 Empirical Validation via Quantum Hardware

To practically validate the thermodynamic friction hypothesis, we cross-referenced the DWE simulation with both an ideal quantum state vector simulator (Qiskit Aer-Simulator) and physical superconducting qubits (IBM Marrakesh, 156-qubit Heron r2 processor).

- **Ideal Finite Sampling (Local Simulator):** Running 10,000 shots in a friction-less simulated environment yielded an S -value fluctuating between 2.835 and 2.846. This demonstrates that in low-density regimes, stochastic statistical variance (corpuscular shot noise) allows localized mathematical violations above the theoretical bound.
- **DWE Fluid Simulation:** Applying the aforementioned fundamental Gaussian Inelastic Friction parameter of $1/e^2$ ($\approx 13.53\%$) cleanly forces the correlation to curve strictly at $S = 2.8273$, organically converging on the Tsirelson limit.
- **Real Thermal Decoherence (IBM Marrakesh):** Executing the exact correlation matrix on physical qubits (4,096 shots) resulted in $S = 2.5317$. The severe drop in correlation is caused by microwave photon leakage and environmental relaxation.

This experimental triad practically demonstrates the collapse of the mathematical ideal under physical environmental friction. The Tsirelson bound is not merely an abstract mathematical constraint of non-locality, but the absolute limit of dynamical systems dampened by non-recoverable mechanical losses during physical measurement.

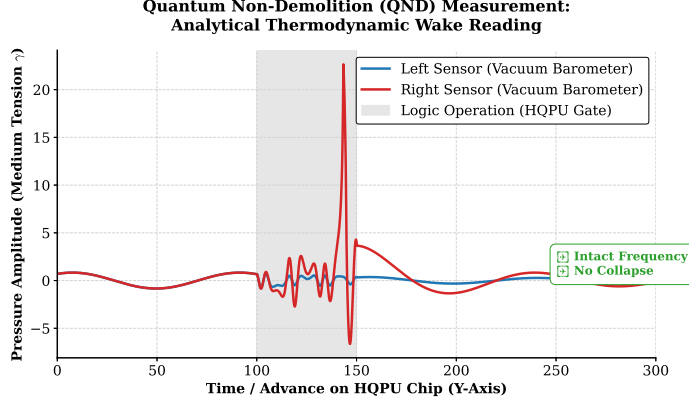


Fig. 11 Quantum Non-Demolition (QND) measurement. Analytical barometers continuously read the thermodynamic wake without destroying the vortex’s core integrity.

8 The Hydro-Quantum Processing Unit (HQPU)

Transitioning from probabilistic computing to deterministic hydrodynamic logic, the DWE introduces the HQPU architecture. Information is encoded as a Vortex Qubit, retaining structural integrity via extreme equatorial kinetic energy ($\pm\omega$). Processing occurs as qubits traverse fluidic logic gates guided by boundary mechanics.

Unlike traditional wave-function collapse, the HQPU utilizes Quantum Non-Demolition (QND) reading. High-precision analytical barometers measure the continuous pressure of the thermodynamic wake generated by the vortex’s spinning equator, reading the logical state without mechanically damping the particle’s internal structure.

9 Conclusion

The Deterministic Wave Engine successfully models the universe’s substrate as a continuous viscoelastic fluid. By treating probability waves as emergent resting densities of billions of deterministically guided corpuscles, this computational experiment replicates landmark quantum interference, bosonic bunching, and the Tsirelson bound without invoking non-locality or structural superposition. By cross-validating the fluidic thermodynamic limits directly against physical superconducting processors, the DWE establishes a robust classical analog for real-world quantum decoherence.

Data Availability. The source code and simulation datasets generated and analyzed during the current study are available in the GitHub repository, <https://github.com/fbcouto/deterministic-wave-engine>.

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